NADA ALJAAFARI

Business & Data Analyst | Saudi Arabia
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PROFESSIONAL SUMMARY

Business and data analyst with an M.S. in Business Analytics (First Honors, KFUPM) and a B.S. in Management Information Systems. Experienced in gathering and documenting requirements, testing systems against them, and presenting recommendations to senior management. Uses SQL, Excel and Python to analyze processes and data, run feasibility and scenario analysis, and drive decisions.

EXPERIENCE

Business Analyst Intern | Saudi Irrigation Organization | Mar 2021 – Jan 2022

- Gathered and validated system requirements with business and technical staff, and documented requirements and quality issues for both audiences.
- Tested internal systems against documented requirements and presented redesign recommendations to senior managers.
- Surveyed employees on website usability and turned the results into documented requirements and visual reports for decision-makers.

Executive Secretary | Al-Kifah Academy | Sep 2022 – Nov 2022

- Prepared reports and process documentation for senior leadership; tracked requirements and flagged process inefficiencies.

ANALYTICS PROJECTS

MAKĀMIN: Financial Fraud Detection (Bootcamp Capstone, team of 4) | *Python, XGBoost, Keras, SHAP, NetworkX, Streamlit*

- Led the modelling on 590,540 real e-commerce transactions (3.5% fraud): compared XGBoost and a neural network, with and without relationship features, using a time-based split to prevent data leakage.
- Final XGBoost model reached 0.521 average precision (15× random) and 0.909 ROC-AUC, catching 74% of fraud; engineered 5 label-free relationship features and used network analysis to expose coordinated fraud (35 linked transactions on one card in 73 hours).
- Translated results for fraud teams through precision–recall threshold trade-offs and SHAP explanations, delivered in MAKĀMIN, a Streamlit dashboard that takes investigators from risk score to reasons to connected transactions.

Emergency Department Analytics (team of 4) | *Python, SARIMA, Random Forest, SMOTE*

- Analyzed 90,036 emergency visits (Royal Commission Hospital, Jubail, 2022) to support staffing decisions: arrivals peak 10 AM–4 PM and the average wait is 33.3 minutes; a SARIMA forecast of daily volume reached 21.3% MAPE vs. 26.6% for Prophet.
- Led the severity classification: showed the 75.4%-accuracy decision tree missed every critical (Severity-1) case, then raised Severity-1 recall from 0 to 0.40 and macro F1 from 0.32 to 0.48 with Random Forest + SMOTE.

Saudi Arabia as a Medical Tourism Hub (Master's Capstone) | *Python, hypothesis testing, classification*

- Compared 79,000 patient records across four hospitals: Saudi and non-Saudi costs differed by only ~1% (SAR 49.8K vs. 50.2K, p = 0.03) and clinical profiles did not separate the groups (59.7% vs. 50% chance); turned the findings into five Vision 2030 policy recommendations.

Marketing Budget Optimization, Mostakan Tea Room (team of 4) | *Excel Solver, linear programming*

- Built a linear-programming model for a client to allocate a SAR 5,000 budget across channels (YouTube 2,351 / TikTok 1,749 / other 300) for an estimated reach of ~1,202.
- Ran sensitivity and feasibility analysis (plan infeasible above SAR 7,823); recommended a moderate budget increase with diversified spend.

SKILLS

Business Analysis: Requirements gathering and documentation, process analysis and documentation, feasibility and scenario analysis, requirements-based system testing, usability testing and UX research, stakeholder reporting and presentations

Data & Reporting: SQL, Microsoft Excel (Solver), Python (pandas, scikit-learn, XGBoost, TensorFlow/Keras, NetworkX), Streamlit, R, PowerPoint, Visio, Jupyter, Git/GitHub

Methods: Fraud detection and imbalanced classification, network analysis, model explainability (SHAP), forecasting, optimization and sensitivity analysis, hypothesis testing

EDUCATION

M.S. Business Analytics, First Honors | King Fahd University of Petroleum & Minerals (KFUPM) | 2024 – 2026

B.S. Management Information Systems | King Faisal University | 2016 – 2021

CERTIFICATIONS & LANGUAGES

Certifications: Data Science Bootcamp, Saudi Digital Academy (Aug – Sep 2026) | Google UX Design Certificate (2024) | UX Design (Accelerated), Misk Skills (2023) | PMP Training (2021) | Commercial Marketing Management Training (2021)

Languages: Arabic: Native | English: Professional working proficiency (IELTS Academic 7.0, CEFR C1)